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RADIATIVE PROCESSES IN EXOPLANET ATMOSPHERES

1 Introduction

The light transport equations can be challenging to solve, in particular when scattering is present, since specific intensities are both position and angle-dependent. Multiple numerical techniques exist to solve the equation, from Monte-Carlo to finite-volume-based methods.

In this project we will solve the equations in the context of planetary atmospheres via a Monte-Carlo algorithm. The aim is to develop an interactive test tool that can be used to visualise exoplanet atmosphere light diffusion emitted by a parent star in a generic context. The tool allows users to visualize images of the sky as seen by an observer on the surface of the planet, as well as incoming spectra. The parameters of observation as well as the composition of the atmosphere and incoming light can be tweaked by the user.

In this stage of the report, we describe the implementation the Monte Carlo procedure in a simplified scenario, considering only scattering.

2 Radiative Transfer

One of the reasons why scattering processes make radiative transfer so challenging is the coupling between light rays at different angles. In this work, we will consider pure scattering, with a coefficient α_ν and differential cross section σ_ν, which are spatially constant inside the atmosphere. The radiative transfer equation along some ray in direction k, parametrized by its length l is as follows:

$$\frac{dI_\nu}{dl}(l,k) = -\alpha_\nu I_\nu(l,k) + \frac{1}{4\pi}\int \sigma_\nu(k \to k')I_\nu(l,k')d\Omega(k') \tag{1}$$

The integral term in the right hand side couples rays from all directions, rendering the equation extremely difficult to solve. Instead, the same equation may be rewritten in the alternative form:

$$\frac{dI_\nu}{d\tau_\nu} = -I_\nu + S_\nu \tag{2}$$

where $\tau_\nu = \alpha_\nu l$ is the optical depth along the ray, and the source function is $S_\nu = \frac{1}{\alpha_\nu}\frac{1}{4\pi}\int \sigma_\nu(k \to k')I_\nu d\Omega$.

In order to solve the equation in the whole space considered, one may decouple the computation of the source function from the formal solution of the equation. This is then performed in two phases, as follows: the first phase consists of a Monte Carlo method, with the purpose of estimating the source function at each point in the atmosphere; the second phase is forward ray tracing, where we integrate the radiative transfer equation using the source computed in the previous step.

In the following sections we describe the algorithm in further details, inspired by Dullemond's Lectures "Radiative transfer in astrophysics"[5].

2.1 Monte-Carlo

As discussed above, the objective of this Monte-Carlo algorithm is to compute the source function at each point in the atmosphere, for an observer placed at some position $s_{obs} \in \mathbb{R}^3$ at the surface of the planet. This observer will receive the rays of light for which we wish to solve the transfer equation.

The main idea of the algorithm is to follow the path of photons in a random-walk as they scatter inside the atmosphere.

We begin by emitting packets of photons from each lightsource in a random manner (at the moment we consider only a single star). Each of these packets will have some random direction and frequency. They will enter the atmosphere and continue travelling in a straight line until the occurrence of a scatter event. During this event, the photon packet will be re-emitted in a new random direction, distributed according to the scattering process considered. It will follow this new direction until the next scatter event, repeating the process until it leaves the atmosphere. At each step of this process, the source function is updated along its "trail".

2.1.1 Grid

The planetary atmosphere is shaped as a spherical shell of thickness H, around the planet (of radius R_p). In practice, it can be modelled as a grid of cells in Cartesian Coordinates, each cell of size $(\Delta X, \Delta Y, \Delta Z)$. We may label each of these cells with a tuple of non-negative integers (i, j, k), such that the center of a cell is placed at position

$$x = \left(i + \frac{1}{2}\right)\Delta X, \quad y = \left(j + \frac{1}{2}\right)\Delta Y, \quad z = \left(k + \frac{1}{2}\right)\Delta Z$$

Figure 1 illustrates the model. Quantities defined inside the atmosphere (e.g. the source function) may be stored inside these cells. Numerically, since $H \ll R_p$, it is preferable to use a sparse array to store such quantities, avoiding the excessive use of storage space[1].

[1] We can estimate the amount of saved space from the ratio between the volume of the spherical shell and cube.

$$\frac{SparseArrayMemory}{DenseArrayMemory} \approx \frac{\frac{4}{3}\pi\left((R_p+H)^3 - R_p^3\right)}{(R_p+H)^3} \sim \frac{4}{3}\pi\frac{H}{R_p}$$

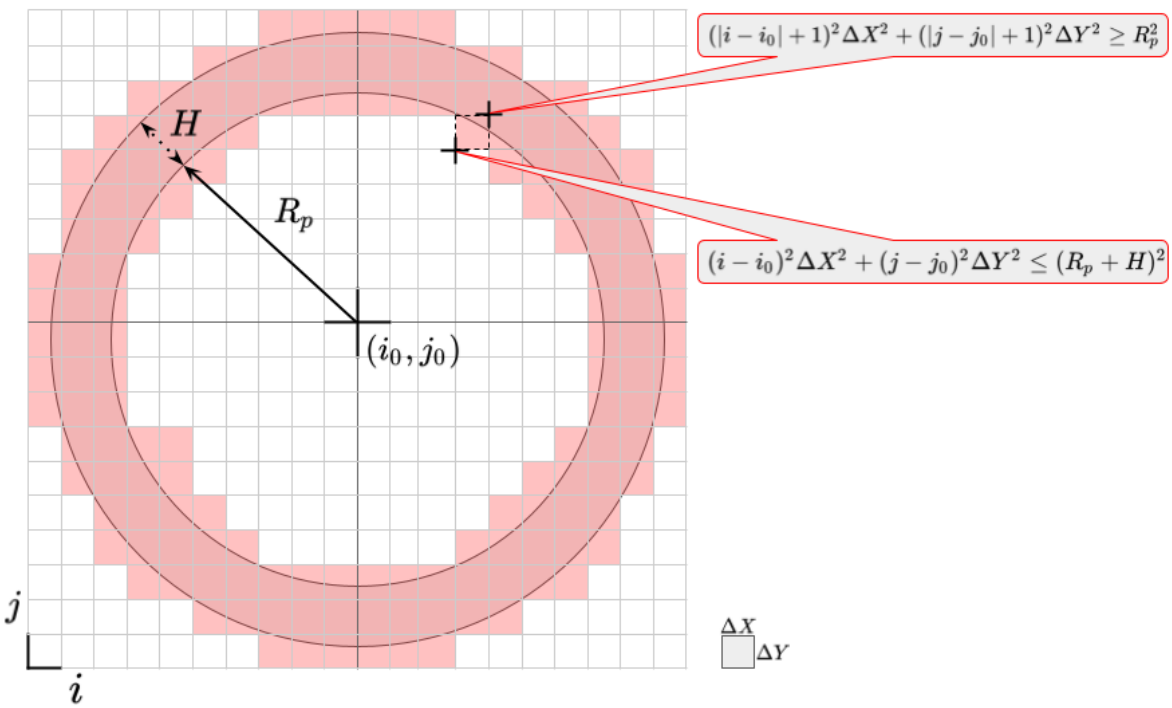

Figure 1: Illustration of grid setup (2D example). Cells that intersect with the atmosphere are shaded in red. The conditions for intersection are schematised for a cell in the first quadrant.

This means that cells completely outside the spherical shell are redundant, and need not be stored in our sparse array. In the diagram, this corresponds to unshaded cells. In practice, this translates to the condition that, if **any** corner of given cell is contained inside the spherical shell, the cell intersects the atmosphere and must be considered. Accounting for symmetries along each axis, the conditions on each cell may be reduced down to two (both of which must be simultaneously true):

$$(i-i_0)^2\Delta X^2 + (j-j_0)^2\Delta Y^2 + (k-k_0)^2\Delta Z^2 \leq (R_p+H)^2$$
$$(|i-i_0|+1)^2\Delta X^2 + (|j-j_0|+1)^2\Delta Y^2 + (|k-k_0|+1)^2\Delta Z^2 \geq R_p^2$$

The constants i_0, j_0, k_0 correspond to the index coordinates of the center of the spherical shell:

$$i_0 = \frac{R_p+H}{\Delta X}, \quad j_0 = \frac{R_p+H}{\Delta Y}, \quad k_0 = \frac{R_p+H}{\Delta Z}$$

Projection to Sphere

Once rays are emitted from a lightsource, they travel through vacuum until eventually hitting the exoplanet atmosphere. The position where the photon packet begins its interaction with the atmosphere must be known.

This can be computed by taking the intersection of the light ray with the sphere of center s_p and radius $R_p + H$. This intersection yields (in the general case) two solutions, from which we must take the closest to the original photon position s. In other words, for a photon emmitted at s, travelling in direction γ:

$$proj(s) = \arg\min \Big\{ r \mapsto ||r-s|| \quad ; \quad r \in Ray \cap Sphere \Big\} \tag{3}$$

where,

$$Ray = \{r \in \mathbb{R}^3 \text{ , s.t. } r = s + \lambda\,\gamma \text{ with } \lambda > 0\}$$
$$Sphere = \{r \in \mathbb{R}^3 \text{ , s.t. } ||r - s_p|| = R_p + H\}$$

We will not enter into much detail on this geometric procedure.

Ray Intersections with Cell Boundaries

As one follows the paths of photon packets, it is necessary to keep track of the current cell and current position of the packet. Since photons move in rectilinear trajectories between two scattering events, an essential geometric computation is to determine when a photon crosses from one cell to the next. This can be viewed as computing the intersection between the ray and cell boundaries.

Suppose the photon packet is currently in cell (i, j, k), at point $P = (P_x, P_y, P_z) \in \mathbb{R}^3$, moving in some direction $\gamma = (\gamma_x, \gamma_y, \gamma_z) \in \mathbb{R}^3$, as illustrated in Figure 2. We may describe its motion by the ray $\Gamma = \{P + \lambda\gamma \quad s.t. \quad \lambda > 0\}$.

Consider the planes:

$$\begin{aligned}
\Pi_i &= \{(x,y,z) \in \mathbb{R}^3 \quad s.t. \quad x = i\Delta X\} \\
\Pi_j &= \{(x,y,z) \in \mathbb{R}^3 \quad s.t. \quad y = j\Delta Y\} \\
\Pi_k &= \{(x,y,z) \in \mathbb{R}^3 \quad s.t. \quad z = k\Delta Z\} \\
\Pi_{i+1} &= \{(x,y,z) \in \mathbb{R}^3 \quad s.t. \quad x = (i+1)\Delta X\} \\
\Pi_{j+1} &= \{(x,y,z) \in \mathbb{R}^3 \quad s.t. \quad y = (j+1)\Delta Y\} \\
\Pi_{k+1} &= \{(x,y,z) \in \mathbb{R}^3 \quad s.t. \quad z = (k+1)\Delta Z\}
\end{aligned}$$

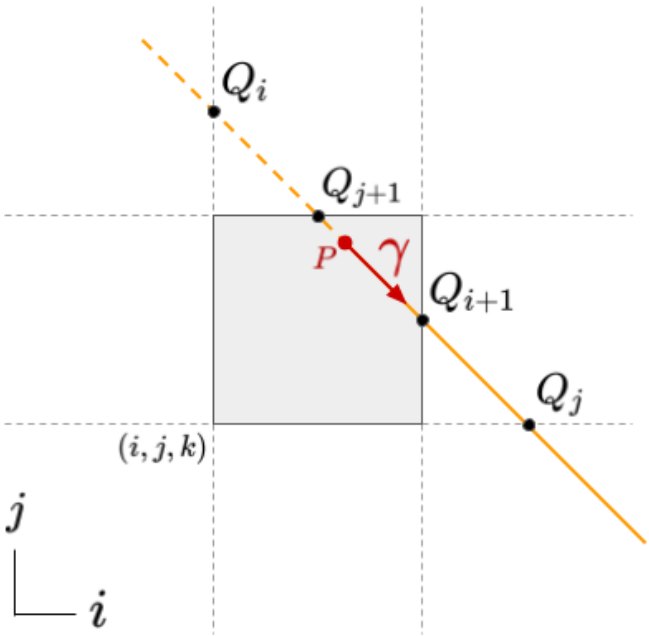

Figure 2: Schematics of some ray crossing cell (i, j, k). Ray starts at some point P, and moves along direction γ. Points of notice Q_i, Q_j, Q_{i+1} and Q_{j+1} are intersections with bounding planes.

The boundaries of the cell are contained in the union of these planes. One may compute the intersections between each plane and the ray, respectively:

$$Q_i = P + \left(\frac{i\Delta X - P_x}{\gamma_x}\right)\gamma; \qquad Q_{i+1} = P + \left(\frac{(i+1)\Delta X - P_x}{\gamma_x}\right)\gamma;$$
$$Q_j = P + \left(\frac{j\Delta Y - P_y}{\gamma_y}\right)\gamma; \qquad Q_{j+1} = P + \left(\frac{(j+1)\Delta Y - P_y}{\gamma_y}\right)\gamma;$$
$$Q_k = P + \left(\frac{k\Delta Z - P_z}{\gamma_z}\right)\gamma; \qquad Q_{k+1} = P + \left(\frac{(k+1)\Delta Z - P_z}{\gamma_z}\right)\gamma;$$

Among these six points, the intersection point will be the closest to P, such that $\lambda > 0$.

2.1.2 Source Function

The source function is a quantity that depends on position, direction and frequency. We may store it in a sparse array in the spirit of Section 2.1.1. Let us name this array "*Source*". For some cell indexed by (i, j, k), and some frequency ν from a discrete set of values, we relate the array *Source* to the source function in this cell:

$$Source[i, j, k, \nu] \equiv S_\nu(s, s_{obs} - s) \tag{4}$$

where s is the position of the cell center, and $(s_{obs} - s)$ is the direction towards the observer. Explicitly:

$$s = \left(\left(i + \frac{1}{2}\right)\Delta X, \left(j + \frac{1}{2}\right)\Delta Y, \left(k + \frac{1}{2}\right)\Delta Z\right)$$

From Equation 4, note that we are only computing the source function at some fixed direction, which corresponds to the direction towards the observer. This means that in a real-time scenario, with a non-static observer, the source function needs to be recomputed at each time-step. Furthermore, for a given cell, we only consider the source function at the cell center. The underlying approximation is that the source function is

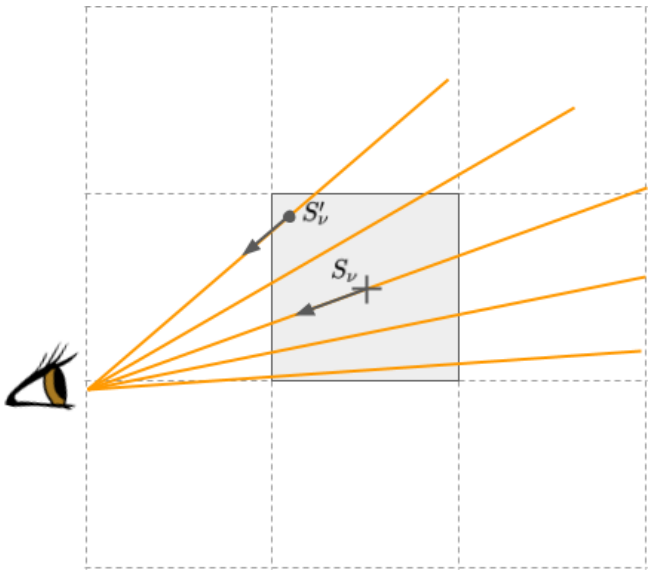

Figure 3: Illustration of the source function at different points in a cell. In the piece-wise constant approximation, $S'_\nu = S_\nu$. Note that the angular argument $(s_{obs} - s)$ and $(s_{obs} - s')$ greatly differ for an observer close to the cell. This may lead to a large approximation error.

constant inside the cell, both spatially and in terms of direction, as illustrated in Figure 3. This may present an issue for cells close to the observer, since a small variation in position leads to a large variation in directional component $(s_{obs} - s)$.

In the Appendix A, we propose a solution to these issues, by storing the source array in terms of function coefficients. This may be implemented in future work.

2.1.3 Photon Emission

The first step of the Monte Carlo simulation consists of emitting a set of $N_\gamma \in \mathbb{N}$ photon packets from each lightsource. Consider the observer placed on a planet centered at $s_p \in \mathbb{R}^3$, and with radius R_p and atmosphere depth H. Consider a stellar lightsource centered at $s_\star \in \mathbb{R}^3$, and with radius $R_\star$. Stars are (approximately) Lambertian emitters, and emit uniformly over their surface. When sampling a random photon packet emitted from the star, we must pick a position, a direction, and a frequency.

The position may be sampled first, uniformly on the stellar surface. However, only rays emitted from one hemisphere of the star may reach the planet. We may thus sample positions uniformly on this hemisphere and consider that this event occurs with probability 1/2. Note that the hemisphere in question has its pole in the direction $(s_p - s_\star)$.

Once the position is sampled, we may sample an emission direction. Consider the spherical coordinates (θ, ϕ), with the polar axis defined at the normal to the emitting surface. Let ϕ be the azimuth and $\mu = \cos(\theta)$ the cosine of the polar angle.

From Dullemond [5] (Eqs 5.37 and 5.38), Lambertian emitters have probability density $f_\mu(x) = 2x$ and can be probabilistically sampled as

$$\begin{aligned}
&\phi \sim Uniform([0, 2\pi]) \\
&\xi \sim Uniform([0, 1]) \\
&\mu = sqrt(\xi)
\end{aligned}$$

However, consider the emission of photons that actually hit the planet. These photons, emitted from the stellar surface, must travel from the star to the planet and thus must have been emitted in a direction within a very small solid angle $\Delta\Omega$. This angle has an aperture of the order $\alpha \approx \dfrac{2(R_p + H)}{||s_p - s_\star||} \ll 1$. As this is small, we may approximate the distribution of directions inside $\Delta\Omega$ as uniform. The probability of a photon being emitted in a direction γ inside this solid angle $\Delta\Omega$ is $\mathcal{P}(\gamma \in \Delta\Omega) = \dfrac{1}{4\pi}\displaystyle\int 2\mu d\Omega \approx \dfrac{\bar{\mu}|\Delta\Omega|}{2\pi}$, where $\bar{\mu}$ is the cosine of the angle between the star surface normal $\hat{n}$ and some "average direction" in $\Delta\Omega$.

Since the planet as seen by the star is a disk, $\Delta\Omega$ is a spherical cap. We must therefore sample a direction uniformly in this spherical cap.

Let us note the cone aperture as α. In spherical coordinates (considering the polar axis to be the same as the cone axis, and $\mu = \cos\theta$), the spherical cap samples as follows:

$$\begin{aligned}
&\phi \sim Uniform([0, 2\pi]) \\
&\mu \sim Uniform([\cos(\alpha/2), 1])
\end{aligned} \tag{5}$$

The solid angle subtended by a spherical cap is $|\Delta\Omega| = 2\pi(1 - \cos(\alpha/2))$.

- Sampling Frequency

The frequency of the emitted photon can be sampled from a Plank function, from a Blackbody with temperature $T_\star$.

We perform the blackbody sampling in a numerical manner. That is, we must first precompute the probability distribution $\mathcal{P}(\nu \leq x)$. Given a partition of frequencies $(\nu_i;\ i = 0, ...N)$, we calculate the Plank function at these values $B_i = B(\nu_i, T_\star)$, then numerically integrate to obtain $F_i = \mathcal{P}(\nu \leq \nu_i)$, and the respective normalized values $\tilde{F}_i = F_i / \sum F_j$. This gives us the cummulative probability distribution $\tilde{F}_\nu(x) = \mathcal{P}(\nu \leq x)$ up to numerical integration error.

Finally, to sample a frequency ν for some photon packet, we first sample a value $\xi \sim Uniform([0, 1])$, then apply the inverse $\nu = \tilde{F}_\nu^{-1}(\xi)$. This can be performed via lookup table in log-time.

- Probability of Hitting the Planet

We may combine the probabilites of the events considered above, i.e., the probability that the photon hits the planet. Firstly, we restricted the position sampling to a hemisphere, with probability 1/2. Next, we sampled directions inside a solid angle. The combined probability of this emission is therefore:

$$\mathcal{P}(Emission) = \mathcal{P}(s \in Hemisphere)\mathcal{P}(\gamma \in \Delta\Omega) = \bar{\mu}\left(\frac{1-\cos\alpha}{2}\right) \tag{6}$$

2.1.4 Scattering Events

Having sampled a photon and followed it to the atmosphere, we must now consider the probabilities associated with each scattering event.

Scattering Distance

Photon packets will travel some random distance inside the atmosphere before being scattered. In terms of optical depth, the distance traveled is on average $\langle\tau\rangle = 1$ (a photon is expected to travel around one unit of optical depth). Furthermore, the probability of travelling some distance $\mathcal{P}(\tau \geq x)$ decays exponentially with this distance[2]. This defines a "memoryless" Poisson process, and can be sampled from an exponential distribution in standard libraries:

$$\Delta\tau \sim Exponential(\lambda = 1) \tag{7}$$

Scattering Direction

After a scattering event, a new photon travelling direction must be sampled. These directions do not follow an isotropic distribution, since Rayleigh scattering is considered. In this case the angles can be sampled from:

$$\begin{aligned} \phi &\sim Uniform([0, 2\pi]) \\ \mu &\sim RayleighPhaseFunction() \end{aligned} \tag{8}$$

where μ is the cosine of the angle between the original and the scattered direction.

The probability density of μ is determined from the Rayleigh Phase Function Φ:

$$f_\mu(x) = \frac{1}{2}\Phi(x) = \frac{3}{4}(1+x^2), \quad -1 \leq x \leq 1$$

We can obtain the cumulative distribution $F_\mu(x) = \mathcal{P}(\mu \leq x)$ by integrating the above in the domain $[-1, 1]$.

$$F_\mu(x) = \frac{1}{4}(x^3 + 3x + 4)$$

To sample μ, one can invert the cumulative distribution:

$$\begin{aligned} \mu &= F_\mu^{-1}(\xi) \\ \xi &\sim Uniform([0, 1]) \end{aligned}$$

Since F_μ is a third-degree polynomial, F_μ^{-1} can be obtained analytically:

$$\begin{aligned} F_\mu^{-1}(\xi) &= g(\xi) - 1/g(\xi) \\ g(\xi) &= \sqrt[3]{\sqrt{16\xi^2 - 16\xi + 5} + 4\xi - 2} \end{aligned} \tag{9}$$

[2] This is described in Dullemond [5] in terms of the probability density $p(\tau) = e^{-\tau}$ (Equation 4.22).

2.1.5 Pseudocode

Algorithm 1 (Monte-Carlo)

For an observer on the surface of a planet of given atmospheric composition, and a parent star, compute an estimate the source function at each point in the atmosphere.

- **Input:**
  - Position of the observer s_{obs}
  - Planet details:
    - Radius R_p
    - Center s_p
  - Atmospheric details:
    - Depth H
    - Number density of particles n
    - Composition (dominant element):
      - Polarizability η (or cross-section $\sigma_s(\nu)$)
  - Stellar details:
    - Temperature $T_\star$
    - Radius $R_\star$
    - Position $s_\star$
- **Output:**
  - Source function at each frequency and each point in the atmosphere, towards the observer $S[\nu, cell]$
- **Parameters:**
  - Number of photon packets N_γ.
  - Grid size Δx, Δy, Δz

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function InitializePhotonPackets

  for each photon $i \in 1..N_\gamma$ **do**

    Sample a frequency from the blackbody distribution:

      $\nu_i \leftarrow SampleBlackbody(T_\star)$

    Sample a position on the star surface (only consider hemisphere pointing towards the planet):

      $s_i \leftarrow SampleHemisphereUniform(center = s_\star, radius = R_\star, axis = s_p - s_\star)$

    Sample a direction (given that it hits the planet):

      view angle: $\alpha \leftarrow \arctan\left(\frac{R_p + H}{||s_p - s_i||}\right)$

$dir_i \leftarrow SampleSphericalCapUniform(axis = s_p - s_i, aperture = 2\alpha)$

Compute the luminosity of the packet:

Calculate monochromatic luminosity of star at frequency ν_i:

$$L_\star^{\nu_i} \leftarrow 4\pi^2 R_\star^2 \times PlankFunction(T_\star, \nu_i)$$

Angle between emission direction and normal:

$$\mu \leftarrow \left\langle \frac{s_i - s_\star}{||s_i - s_\star||}, dir_i \right\rangle$$

Probability of emitting photon from hemisphere and within spherical cap:

$$\mathcal{P}(Emission) \leftarrow \mu \left(\frac{1 - \cos\alpha}{2} \right)$$

Weigh the packet luminosity by the probabilities of emission and number of photon packets:

$$l_i \leftarrow \left(\frac{1}{N_\gamma} \mathcal{P}(Emission) \right) L_\star^{\nu_i}$$

Follow the photon in the vacuum until it hits the atmosphere.

$s_i \leftarrow ProjectToSphere(s_i, center = s_p, radius = R_p + H)$

end for

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function RunMonteCarlo

for each photon $i \in 1..N_\gamma$ **do**

while s_i is inside atmosphere **do**

$\Delta\tau \leftarrow SampleExponential(\lambda = 1)$

while $\Delta\tau > 0$ **do**

Compute scatter coefficient and cross-section, in current cell:

$$\sigma_s \leftarrow \frac{8\pi}{3} \left(\frac{2\pi\nu_i}{c} \right)^4 \eta^2$$

$$\alpha_s \leftarrow n\sigma_s$$

Get the intersection point where the photon reaches the end of current cell:

$bound \leftarrow IntersectionOfRayAndCell(s_i, dir_i)$

Calculate how much the photon will move in the current cell (the photon will either diffuse inside the cell, or reach the boundary to the next cell):

$\Delta s \leftarrow min(||bound - s_i||, \Delta\tau\alpha_s)$

Update source function:

Compute the angle (cosine) between the photon direction and the observer direction:

$$\mu \leftarrow \left\langle \frac{s_{obs} - s_i}{||s_{obs} - s_i||}, dir_i \right\rangle$$

Compute the value of the Rayleigh Phase Function at μ

$\Phi \leftarrow RayleighPhaseFunction(\mu)$

Compute the volume of the cell:

$\Delta V \leftarrow \Delta X \Delta Y \Delta Z$

$\Delta Source \leftarrow \frac{l_i \, ds \, \alpha_s}{4\pi\Delta V} 2\Phi$

$Source[\nu_i, cell] \leftarrow Source[\nu_i, cell] + \Delta Source$ ▷ *cell* is the current cell

Move the photon along its trajectory:

$s_i \leftarrow s_i + \Delta s \, dir_i$

$\Delta\tau \leftarrow \Delta\tau - \Delta s/\alpha_s$

end while

Once $\Delta\tau = 0$, diffusion will occur. Set a new scattered direction:

$dir_i \leftarrow SampleRayleighPhaseFunction()$

end while

end for

2.2 Raytracing

Once the source function is computed, the next step consists of integrating the radiative transfer equation to obtain the specific intensity at different frequencies, in the observer's position. This can be done by drawing rays from the camera, from each pixel, going through the focal plane, and traversing the atmosphere. These rays correspond to the locus where we must integrate to obtain $I_\nu(\tau_{obs})$.

The initial value $I_\nu(0)$ for the specific intensity will depend on the direction of the ray. If the ray hits the star, then this intensity will be the Plank function at that frequency. If not, it will be zero.

The source function at each cell i in the ray has already been calculated with the MonteCarlo simulation. Therefore, we can obtain the intensity at the observer's position by doing piecewise integration of the radiative transfer equation:

$$I_\nu(\tau_{obs}) = I_\nu(0)e^{-\tau_{obs}} + \sum_i S_i(e^{\tau_i} - e^{\tau_{i+1}})$$

The above equation can be easily derived from the transfer equation by considering the source function at each cell to be a constant.

2.3 Coloured Images

At this point we have obtained specific intensities at the observer's position for a set of frequencies in the domain defined by the Monte-Carlo simulation. Using the method described in the previous report we can obtain the flux densities F_ν.

From this, we may wish to not only obtain an integrated flux over all frequencies, but instead construct an RGB image. One way to do this is by following the CIE Color Standard[6]. This standard was defined from empirical testing on the human eyesight and is widely used to transform spectral data into an RGB basis.

Human eyesight has a great degeneracy of colours. Different spectra may be interpreted as the same colour. This occurs because human "cones" (receptor cells responsible for detecting coloured light) are activated by specific bands of frequencies; in practise, cones can be modeled as filters that act on the spectral signal. One can define three basis functions g_R, g_G and g_B corresponding to three types of cones that yield "intensities" of red, green and blue (Figure 4). This filters then act on the incoming spectrum (specific intensity) to yield three values at each pixel: R, G and B.

Filters act by convolution at the origin:

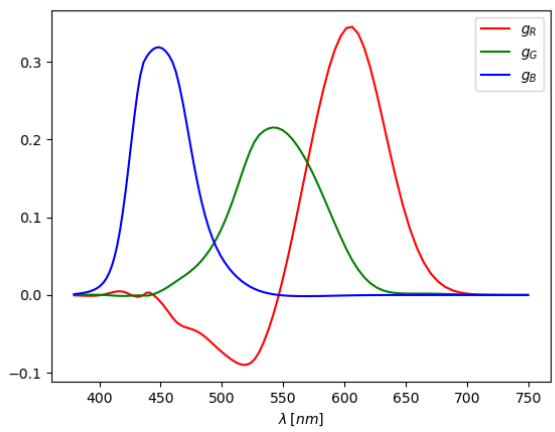


Figure 4: RGB filter basis.

$$R = \int I_\nu g_R(\nu) d\nu$$
$$G = \int I_\nu g_G(\nu) d\nu$$
$$B = \int I_\nu g_B(\nu) d\nu$$

The basis functions are defined according to CIE Standard. Firstly, define the XYZ-space basis functions g_X, g_Y, g_Z:

$$g_X(\lambda) = 1.056 g(\lambda, 599.8, 37.9, 31.0) + 0.362 g(\lambda, 442.0, 16.0, 26.7) - 0.065 g(\lambda, 501.1, 20.4, 26.2)$$
$$g_Y(\lambda) = 0.821 g(\lambda, 568.8, 46.9, 40.5) + 0.286 g(\lambda, 530.9, 16.3, 31.1)$$
$$g_Z(\lambda) = 1.217 g(\lambda, 437.0, 11.8, 36.0) + 0.681 g(\lambda, 459.0, 26.0, 13.8)$$

Here, $g(\lambda, m, a, b)$ correspond to piecewise gaussians, with λ the wavelength:

$$g(\lambda, m, a, b) = \begin{cases} e^{-0.5(\lambda - m)^2/a^2} & \text{if } \lambda < m \\ e^{-0.(\lambda - m)^2/b^2} & \text{if } \lambda \geq m \end{cases}$$

Finally, this basis can be transformed from XYZ to RBG by the operation:

$$\begin{pmatrix} g_R \\ g_G \\ g_B \end{pmatrix} = 0.17697 \begin{pmatrix} 0.49000 & 0.31000 & 0.20000 \\ 0.17697 & 0.81240 & 0.01063 \\ 0.00000 & 0.01000 & 0.99000 \end{pmatrix}^{-1} \begin{pmatrix} g_X \\ g_Y \\ g_Z \end{pmatrix}$$

The obtained scalar numbers R, G, B correspond to the red, green and blue channel of a given pixel, for specific intensity I_ν computed at this pixel.

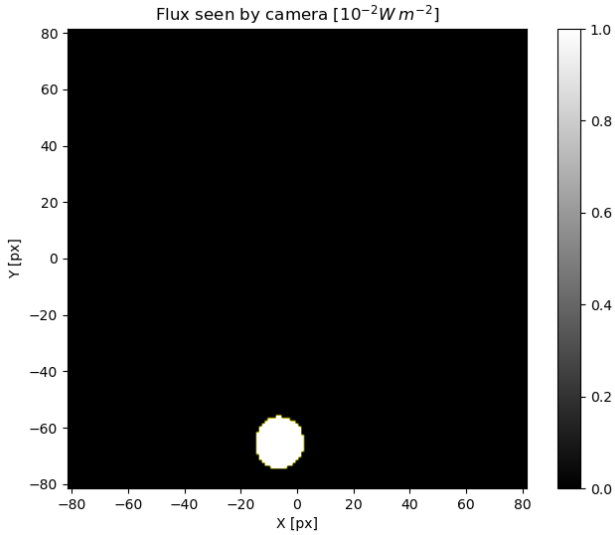


Figure 5: Rendering of the star on canvas, without considering scattering effects.

3 Discussion

In this Section we present analyses of different aspects of the numerical simulations.

3.1 Star

We would wish to verify whether the star, as seen by the observer, presents the expected angular size. For a star with a radius of $10R_\odot = 6.957 \times 10^6$km, distant $1AU = 1.496 \times 10^8$km away from the planet, we have:

$$\Delta\theta = \frac{2R_\star}{||s_{obs} - s_\star||} \approx 5.32^\circ$$

This is ten times the angular size of the sun (as expected for a star with $10R_\odot$). Its diameter when projected to canvas is (supposing aligned with the optical axis):

$$\Delta x = \frac{2R_\star}{||s_{obs} - s_\star||} f_0 = 3.26 \text{ mm} = 16.3 \text{ px}$$

for a camera of focal length $f_0 = 35$mm and pixel size $px = 200$ microns.

We estimate from Figure 5 that the projected star has a diameter between 15 and 20 px, agreeing with our calculations. Small distortions perceived on the vertical length are due to projection effects on the flat canvas. These distortions increase the further one is from the optical center.

3.2 Atmosphere

We have modelled the atmosphere as a spherical shell of thickness $H = 12$km, around the planet (radius $R_p = 6371$km), with homogenious density and composition, of a single molecule, N_2. Due to the size of the nitrogen molecules, we may suppose pure Rayleigh Scattering:

$$x = 2\pi a/\lambda \ll 1,$$

since we know that $a = 364$pm (kinetic diameter of N_2 [2]) and $\lambda \in [380, 700]$nm (wavelength interval of visible light).

The number density of particles in Earth's **troposphere** at sea level is of the order of $n = 245.657 \times 10^{23}$ particles / m^3, with a thickness of $H \sim 12km$ [3]. The Rayleigh cross-section of N_2 at 532 nm (green light) is $5.1 \times 10^{-31} m^2$ [4]. From this we may calulate the cross-section at other wavelengths $\sigma_s = 5.1 \times 10^{-31} \times \left(\frac{532nm}{\lambda}\right)^4$.

Note that for green light we will have a (perpendicular) optical depth of $\Delta\tau = H\alpha_s = Hn\sigma_s \approx 0.15$. Therefore we can expect light to scatter less than once on average.

3.3 Ballpark Estimates

We can estimate the expected orders of magnitude of the source function and observed specific intensity by considering a very simple case. Let us treat the star as a distant point-like source, since $R_p, R_\star \ll ||s_p - s_\star||$. Also suppose the atmosphere is physically thin with $H \ll R_p$, which means we can approximate its curvature as flat (at least in the region close to the observer). The full setup is illustrated in Figure 6, considering the star as an overhead source.

We define t_1 as the distance from a point P to the observer, and t_2 as the distance from P to the edge of the atmosphere, along the ray that joins P to the lightsource. This is also illustrated in Figure 6.

In the absence of scattering, the mean intensity at a given point would be related to the specific intensity and distance d to the star[3]:

$$J_\nu^0 = B_\nu(T_\star)\left(\frac{1-\cos\theta_c}{2}\right) \sim \frac{B_\nu(T_\star)}{4}\left(\frac{R_\star}{d}\right)^2$$

Inputing the correct values, with $T_\star = 5778$, yields $J_\nu^0 \sim 1.1 \times 10^{-11} W sr^{-1} m^{-2} Hz^{-1}$ in the visible ($B_\nu(T_\star) \sim 2 \times 10^{-8} W sr^{-1} m^{-2} Hz^{-1}$).

[3]Calculated during the first practical session of the course, for a spherical lightsource.

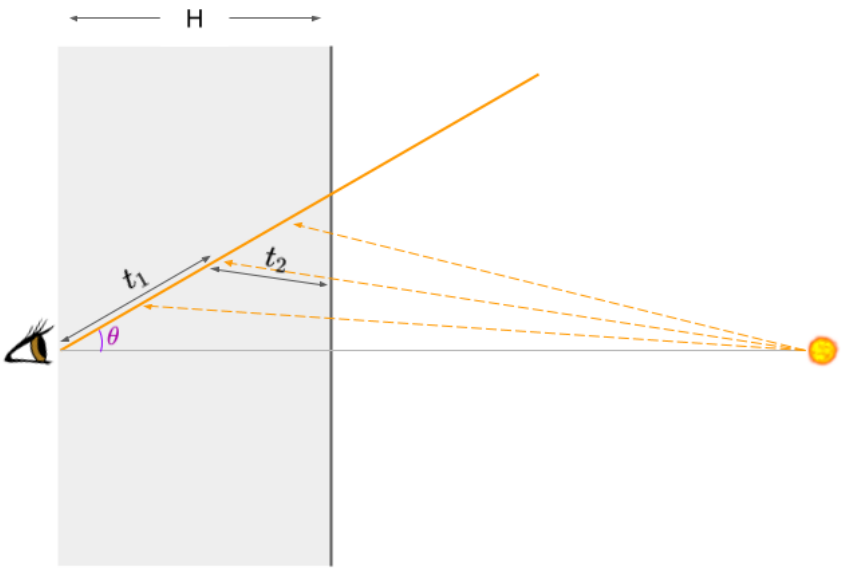


Figure 6: Illustration of the single-scattering approximation, on a planar-like atmosphere. The source function at each point in a ray can be computed from direct rays joining the star and that point.

Instead, we put ourselves in the single scattering regime, where $\Delta\tau = \alpha_\nu H \ll 1$. This means we suppose that light scatters at most once before reaching the observer. In this optically thin regime, we may expect the mean intensity to decay with optical depth:

$$J_\nu = J_\nu^0 e^{-\tau_2}$$

Here, $\tau_2 \equiv \alpha_\nu t_2$ is the optical distance along t_2. In the distant-lightsource approximation, we have the geometrical identity $t_2 \approx H - t_1 \cos\theta$. Rewriting in terms of $\tau_1 \equiv \alpha_\nu t_1$,

$$J_\nu = J_\nu^0 e^{-\alpha_\nu H + \tau_1}$$

We are now ready to integrate the formal radiative transfer equation. To simplify calculations, let us consider isotropic scattering, in which case the source S_ν is equal to the average intensity field J_ν. For an observer at the planet surface, we find the solution:

$$\begin{aligned}
I_\nu^{obs} &= I_\nu^0 e^{-\alpha_\nu H} + \int S_\nu e^{-\tau_1} d\tau_1 \\
&= \left[I_\nu^0 + J_\nu^0 \int_{[0,\alpha_\nu H]} e^{-(\tau_1(1-\cos\theta))} d\tau_1 \right] e^{-\alpha_\nu H} \\
&= \left[I_\nu^0 + J_\nu^0 \left(\frac{1 - e^{-(\alpha_\nu H(1-\cos\theta))}}{1-\cos\theta} \right) \right] e^{-\alpha_\nu H} \\
&\approx \left[I_\nu^0 + J_\nu^0 \alpha_\nu H \right] e^{-\alpha_\nu H}
\end{aligned}$$

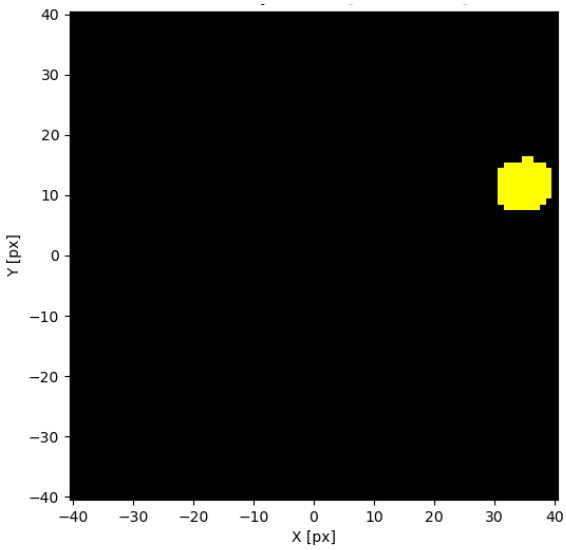
For a ray that leaves the lightsource towards the observer, we have $I_\nu^0 = B_\nu(T_\star)$, whereas for a ray that does not intersect the lightsource, $I_\nu^0 = 0$. Plugging in the values for I_ν^0, J_ν^0 and $\alpha_\nu H \approx 0.15$ (for green light), we can estimate I_ν^{obs} to be 1.7×10^{-8} and 1.4×10^{-12} in these two cases respectively (SI units).

3.4 Monte Carlo Simulations

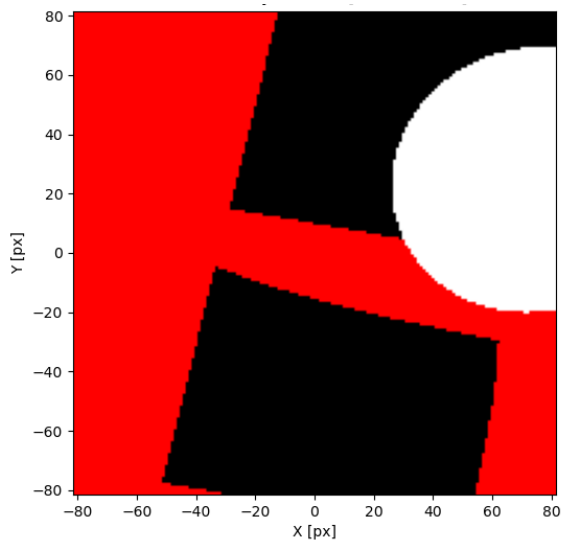
The code was implemented in python, using vectorized numpy and sparse scipy arrays. The simulation was executed in an 11th Gen Intel(R) Core(TM) i7-11850H @ 2.50GHz. The atmosphere grid was subdivided into $(500, 100, 100)$ cells along the (x, y, z) axes respectively. The star is placed on the x axis. The Monte Carlo was run using a total of 100 000 photon packets, aimed towards the observer with an aperture reduced by a factor of a thousand, i.e. $\alpha' = \alpha/1000 = 2\arctan\left(\frac{R_p+H}{||s_p - s_\star||}\right)/1000$ (see Section 2.1.3). This allows us to populate the source function in a small region of the atmosphere with a limited amount of photons. For these parameters, the Monte Carlo simulation took an approximate 1h CPU time.

We observe a specific intensity on the order of 10^{-8} (SI) for rays incoming directly from the star, which agrees with ballpark calculations. However, the observed source function peaks at $\max S_\nu \approx 1 \times 10^{-17}$ (SI), which is a lot lower than expected. We could not determine the reason for this issue. In Figure 7(a) we present a resulting rendered image.

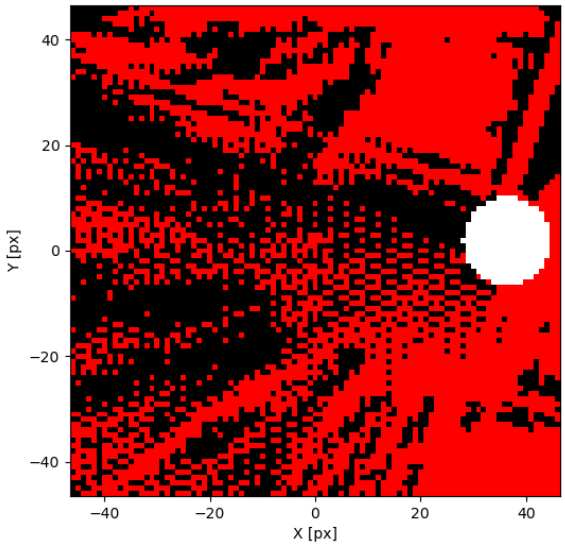
By increasing the brightness scale of the image, it is possible to visualize the scattered source function. We see this in Figure 7(b-d), where we vary different parameters of the problem ($R_\star$, d, α, etc). In each of the images we notice the presence of some artifacts and unexpected behaviour. In particular, we mostly perceive scattering from red light. These indicate hidden issues with the code. The most likely scenario is that they are stem from the implementation of the geometric computations as well as possible numerical approximation issues.



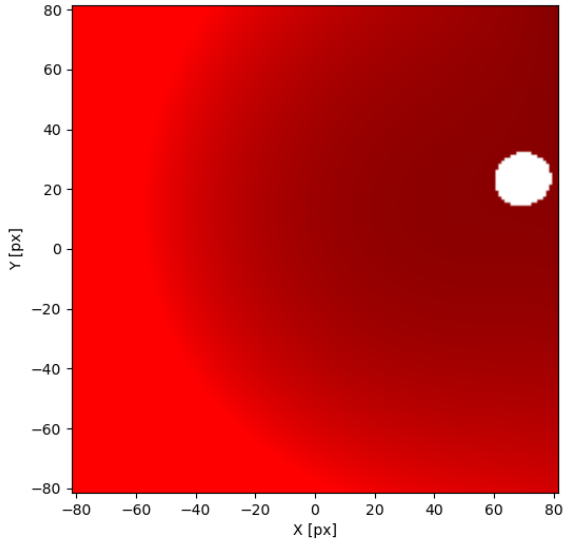
(a)



(b)



(c)



(d)

Figure 7: (a) Flux seen by camera for original simulation with 100000 photon packets. The scattered light is not visible in this rendering. (b-d) Three different renderings were obtained by increasing the brightness scale of the image and varying the parameters: star size, distance, camera view angle and number of photon packets. The presence of artifacts is clearly observed. Additionally, we only observe the scattering of red light. All of these signs indicate issues with the code.

A Appendix: Functional Representation of the Source

In this appendix we propose an alternative way to represent the source function during the Monte-Carlo computation. In the original algorithm, the values of the source at each cell are stored for a fixed observer (i.e., as a scalar). The proposed alternative representation allows one to instead describe S_ν as a **function** (of its angular argument), avoiding the approximation issues mentioned in Section 2.1.2, as well as allowing one to execute the Monte Carlo only **once** for a non-static observer.

Consider some fixed frequency ν. The source $S_\nu : \mathbb{R}^3 \times [0, 2\pi]$ is a function of position and direction. From the Monte Carlo procedure, we have, for some position $s \in \mathbb{R}^3$ and direction $n \in \mathbb{R}^3$:

$$S_\nu(s, n) = \sum_i \frac{l_i \alpha_s ds}{2\pi \Delta V} \Phi(\mu_i) \tag{10}$$

where the sum in i corresponds to different photon packets.

The phase function for Rayleigh Scattering is a second degree polynomial on μ, the cosine of the angle between the photon travel direction γ and n:

$$\Phi(\mu) = \frac{3}{4}(1 + \mu^2)$$

We may rewrite μ in terms of a dot product between n and γ:

$$\mu = \cos(\theta) = n \cdot \gamma$$

With this, μ^2 may be described by a bilinear operator applied on (n, n):

$$\mu^2 = (n \cdot \gamma)^2 = n^T \gamma \gamma^T n \tag{11}$$

Equation 11 allows us to commute the sum in S_ν:

$$\begin{aligned} S_\nu(s, n) &= \sum_i a_i (1 + \mu_i^2) \\ &= \sum_i a_i + \sum_i a_i \left(n^T \gamma_i \gamma_i^T n \right) \\ &= \sum_i a_i + n^T \left(\sum_i a_i \gamma_i \gamma_i^T \right) n \\ &\equiv b_\nu + n^T A_\nu n \end{aligned} \tag{12}$$

From this we can see that S_ν is a quadratic function in n. One may simply store the values b_ν and $A_\nu \in \mathcal{M}_{3\times3}(\mathbb{R})$ for each cell, updating them for new photon packets during the Monte Carlo algorithm. Since A_ν is symmetric, this occupies the memory space of 7 floating points for each given cell (for each frequency value stored), instead of a single floating point. In exchange, one gains better a approximation for S_ν and the possibility of real-time rendering (by storing the source function after a single execution of the Monte Carlo and rendering from a moving viewpoint).

The method presented here can naturally be extended to other phase functions, by rewriting arbitrary powers of μ as k-linear forms (i.e. as tensors). Note however that the number of components that need to be stored in memory will grow as $O(k^2)$[4], where k is the degree of μ^k.

[4] Consider a fully symmetric tensor of order k with indices taking values in $\{1, 2, 3\}$. Such tensor has 3^k components. But after accounting for symmetry, it can be shown that there are in reality only $\frac{(k+1)(k+2)}{2}$ degrees of freedom.

References

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[2] https://en.wikipedia.org/wiki/Kinetic_diameter#List_of_diameters

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